

Intelligent EV Charging Demand & Agentic Infrastructure Planning

This system uses LangGraph-based stateful agent workflow.

Upload charging data → Predict demand → Detect hotspots → Retrieve planning guidance → Reason over optimization trade-offs → Generate an explainable infrastructure plan

System Architecture

- CSV Input
- > Preprocessing
 - > ML Model
 - > Forecast
 - > LangGraph Agent
 - > RAG
 - > Planning Engine
 - > UI

Execution Workflow

- Input Node
- > Preprocessing Node
 - > Prediction Node
 - > Hotspot Detection Node
 - > RAG Retrieval Node
 - > Reasoning Node
 - > Planning Node
 - > Output Node

Input Specification

- CSV file containing station_id, timestamp/hour, and model-ready demand features.
- Minimum operational fields expected by the app: station_encoded and hour.
- Optional trend and calendar features can be imputed when partially missing.

Output Specification

- predictions
- avg_demand
- peak_demand
- peak_hour
- hotspots
- infrastructure_plan
- schedule

Model Performance

MAE

4.13

RMSE

6.08

R² Score

0.690

Model explains approximately 69.0% of the variance in held-out hourly demand with an RMSE of 6.08 kWh.

Evaluated on 2722 held-out rows using chronological split from 2020-02-20 17:00:00.

Upload CSV Dataset



Drag and drop file here

Limit 200MB per file • CSV

Browse files



model_ready_hourly_data.csv 2.3MB



File uploaded successfully.

Select Station

1



Run LangGraph Planning Agent

Executive Summary

Avg Demand

Peak Demand

Peak Hour

Risk

Executive Summary

Avg Demand

4.89

Peak Demand

7.08

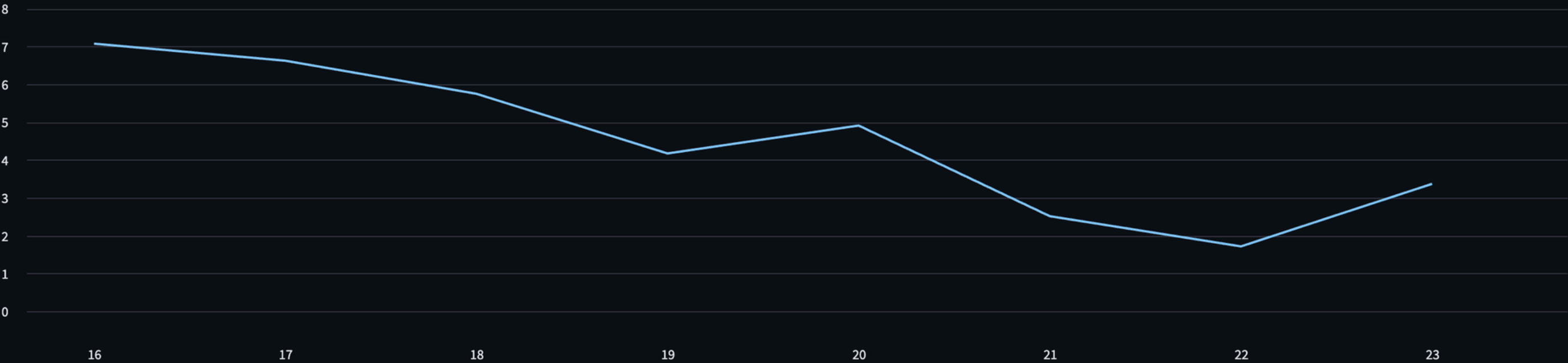
Peak Hour

16:00

Risk

Low

Demand Forecast



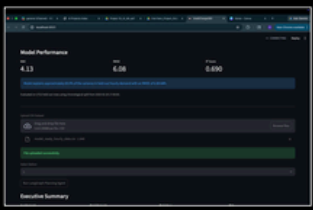
Hotspot Detection

	hour	demand	severity
0	16:00	7.08	High
1	17:00	6.63	High

Infrastructure Plan

	station_id	chargers	charger_type	utilization_target	load_balancing	cost
0	1	2	Level 2 AC Charger	59.0%	Not required	Low

Scheduling Strategy



1	17:00	6.63	High
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Reasoning

- Peak demand is 7.08 kWh at 16:00 with average demand 4.89 kWh.
- Selected 2 Level 2 AC Charger because peak-time utilization is 59.0%, which stays near the 70-90% target band.
- Using fewer chargers would push utilization above 117.9% and risk queuing or overload.
- Adding more chargers would drop utilization to 39.3%, increasing idle infrastructure cost.
- Peak load smoothing target is 6.01 kWh, so scheduling should shift discretionary demand away from the highest-load window.
- Grid load remains within normal operating threshold.

Planning Explanation

The plan chooses 2 chargers instead of a simple peak/10 rule because it optimizes for 70-90% utilization, manages grid risk, and avoids overspending on idle infrastructure. Peak demand is served with 59.0% expected utilization.

Retrieved Guidance

- 2. Load Balancing and Grid Impact:
 - Threshold Management: Sites forecasting peak energy demand over 25 kWh per hour must incorporate load balancing hardware or software to prevent local transformer overload.

Scheduling Strategy

- Peak window (15:00, 16:00, 17:00): prioritize queue control and defer flexible charging.
- Off-peak window (22:00, 21:00, 23:00, 19:00, 20:00, 18:00): incentivize discounted charging and background fleet charging.
- Apply smart scheduling to shift 10-15% of discretionary demand from peak to off-peak hours.

Recommendations

Deploy 2 Level 2 AC Charger units for balanced utilization and acceptable queue risk.

Keep monitoring transformer loading; dynamic balancing can remain on standby.

Use time-of-use pricing and reservation nudges to smooth demand before the peak hour.



The plan chooses 2 chargers instead of a simple peak/10 rule because it optimizes for 70-90% utilization, manages grid risk, and avoids overspending on idle infrastructure. Peak demand is served with 59.0% expected utilization.

Retrieved Guidance

- 2. Load Balancing and Grid Impact:
 - Threshold Management: Sites forecasting peak energy demand over 25 kWh per hour must incorporate load balancing hardware or software to prevent local transformer overload.
 - Time-of-Use Strategies: Stations detecting consistent high demand during peak grid hours (17:00 - 21:00) should implement dynamic pricing to shift loads or cap maximum power output by 20% during these periods.
- EV Infrastructure Planning and Deployment Guidelines

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Structured Output

```
{
  "workflow": "This system uses LangGraph-based stateful agent workflow."
  "raw_data": [
    0: {
      "stationID": "2-39-123-557"
      "hour_timestamp": "2019-07-09 16:00:00"
      "total_kWh": 6.569
      "hour": 16
      "dayofweek": 1
      "month": 7
      "day": 9
      "weekofyear": 28
      "hour_sin": -0.8660254037844384
      "hour_cos": -0.5000000000000004
      "dow_sin": 0.7818314824680298
    }
  ]
}
```

